ME403 Student Simulation Pack Guide

ME403 Introduction to Robotics

1 Purpose

This document explains how to use the released `ME403_LabFiles` package as a student. It clarifies what is included, what is intentionally excluded, and how simulation assets are prepared at runtime.

2 What Is Included

- Lab starter folders under `labs/`
- Vendored SDK and robot description sources under `vendor/`
- Full simulation runtime under `simulation/runtime/`
- Optional simulation demos under `simulation/student/`
- Optional robot setup scripts under `robots/`
- Bootstrap and asset helpers under `scripts/`
- Setup and troubleshooting docs under `docs/`

3 What Is Not Included by Default

- Instructor/private solution material
- Internal course publishing helpers
- Build artifacts and temporary caches

4 Quickstart

Run from package root:

```
cd ME403_LabFiles
python3 -m pip install -r requirements/base.txt
python3 scripts/bootstrap.py --simulation
cd labs/lab01_fk
python3 interface.py
```

5 Simulation Architecture (Student View)

Upstream Assets vs Runtime Assets

The `vendor/` folder stores upstream robot descriptions and SDK sources. When URDF backends run, `simulation/runtime/urdf_loader.py` prepares runtime copies for simulator compatibility.

Preparation and Cache Pipeline

The runtime preparation step can include:

- Mesh format conversion for backend compatibility
- URI/path rewriting so files resolve correctly
- Simulator-specific adjustments and hidden parity chain injection
- Cached output placement under `$DOBOT_SIM_CACHE`

Default cache path is:

```
~/.cache/dobot_sim
```

6 Environment Variables

Variable	Meaning
`DOBOT_SIMULATION`	Set to `1` to force simulation mode (default for labs).
`DOBOT_SIM_BACKEND`	Choose `mujoco` (default) or `pybullet`.
`DOBOT_VIZ`	Set to `0` for headless mode (no GUI windows).
`DOBOT_ROBOT_TYPE`	Choose `magician` or `mg400`.
`DOBOT_EE`	Magician TCP mode: `none` (default, bare flange), `motor`, or `suction`. Controls which TCP offset the sim uses. Must match the real robot's `set_tool_offset()` setting for FK parity.
`DOBOT_SIM_CACHE`	Override cache directory for prepared assets.
`DOBOT_MAGICIAN_URDF_PATH`	Override Magician URDF source file/folder path.
`DOBOT_MG400_URDF_PATH`	Override MG400 URDF source file/folder path.

7 Backend Choice

- Use `mujoco` for the default course simulation workflow.
- Use `pybullet` when MuJoCo graphics/drivers are unavailable.
- Use headless mode (`DOBOT_VIZ=0`) for remote sessions.

8 Troubleshooting Quick Chart

Symptom	Action
Missing URDF/meshes	Run `python3 scripts/fetch_assets.py -all` from package root.
First run is very slow	Expected behavior: initial fetch + mesh prep + cache generation.
MG400 URDF generation error	Install `xacro`, then rerun MG400 asset fetch.
GUI or display issues	Try `DOBOT_VIZ=0`; on Linux test `MUJOCO_GL=egl`.
Stale cache behavior	Delete `$DOBOT_SIM_CACHE` (or `~/.cache/dobot_sim`) and rerun.

9 Matching Sim to Real-Robot EE Mode (Magician)

The simulation TCP mode (`DOBOT_EE`) and the real robot's stored offset must agree for FK/IK errors to reflect kinematics rather than an offset mismatch.

Sim `DOBOT_EE`	**TCP offset (mm)**	**Real robot Python command**
`none` (default)	(0, 0, 0)	`set_tool_offset(bot, 0, 0, 0)`
`motor`	(60, 0, 0)	`set_tool_offset(bot, 60, 0, 0)`
`suction`	(60, 0, −70)	`set_tool_offset(bot, 60, 0, -70)` — physical cup tip

`MAGICIAN_TOOL_OFFSETS` dict in `utils.py` provides shortcuts:

```
set_tool_offset(bot, *MAGICIAN_TOOL_OFFSETS["none"])    # FK labs default
```

For MG400 labs, no TCP offset switching is needed — the MG400 always uses its bare wrist reference point in ME403 labs.

10 Recommended Student Workflow

1. Start in simulation and validate your lab code.
2. Switch robot type/backend only when needed.
3. Move to hardware only after simulation results are stable.
4. For Magician: home once in DobotStudio (v1.9.4), then use `set_tool_offset()` from Python to set the TCP mode for each session.
5. Verify `DOBOT_EE` (sim) matches `get_tool_offset(bot)` (real) before collecting FK/IK data.

11 Further Reading

- `README.md` (root quickstart)
- `docs/simulation.md`
- `docs/troubleshooting.md`